EALAI: An Explainable and Auditable Legal AI Framework with Empirical Evaluation on Singapore Statutory Law

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Abstract. Legal AI chatbots built on large language models frequently hallucinate statutory references and provide no way to trace how an answer was produced. This paper presents EALAI, a legal advisory system that combines hybrid retrieval-augmented generation with a 20-rule symbolic consistency engine and SHA-256 tamper-evident audit logging. The system retrieves from a corpus of 48 Singapore statutory PDFs using Reciprocal Rank Fusion over BM25 and dense retrieval, then cross-checks every LLM response against domain-specific legal rules before logging the full interaction.

We evaluate EALAI on a 200-question benchmark across three experiments. Dense retrieval reduces hallucination from 42% (no-retrieval baseline) to 20% and improves citation accuracy from 0.580 to 0.800. Hybrid retrieval achieves the best raw recall (Hit@5 = 72.5%) but unexpectedly underperforms on end-to-end citation accuracy (0.645), suggesting that context coherence matters more than breadth for LLM attribution. A refusal calibration sweep over ten thresholds identifies $\theta = 0.45$ as the optimal operating point with 3% false positives.

Keywords: Legal AI · Retrieval-Augmented Generation · Explainable AI · Symbolic Reasoning · Audit Trails · Singapore Law

1 Introduction

AI-powered legal chatbots are increasingly used for document review, compliance checks, and basic legal guidance [4]. The core risk is that language models generate plausible but inaccurate text. When a system cites the wrong statute or invents a section number, the person relying on that advice may face real consequences [1, 8]. Tools like DoNotPay have demonstrated both the potential and the limitations of automated legal assistance [3].

Three failure modes recur in existing systems: (1) hallucination of statutory content, (2) absence of citation provenance, and (3) unauditable reasoning incompatible with emerging governance requirements such as the EU AI Act [6] and digital ethics frameworks [5]. Most production systems treat explainability as an afterthought rather than an architectural requirement [7].

This paper presents EALAI, a working system that addresses all three failure modes. Our contributions are:

1. A hybrid RAG pipeline combining BM25 [11] and dense retrieval via Reciprocal Rank Fusion [12], with a 20-rule symbolic consistency engine that cross-checks LLM output against known Singapore legal rules.
2. A SHA-256 tamper-evident audit logger that records every interaction with deterministic hashing for post-hoc verification.
3. Empirical evaluation on a 200-question benchmark showing that RAG reduces hallucination by 22 percentage points, along with the finding that retrieval coverage and end-to-end citation accuracy are not positively correlated.

2 Related Work

Rule-based legal AI. Early systems such as Taxman (McCarty, 1977) encoded legal reasoning as IF-THEN chains. These produced auditable reasoning but could not scale beyond narrow domains [2].

Neural approaches. Transformer models (BERT through GPT-4) enabled broad legal NLP without explicit rules. Domain-specific efforts include DISC-LawLLM [14] for Chinese legal services and the Indian Legal Documents Corpus [15] for court judgment prediction. Commercial products such as Harvey AI and CoCounsel have reached production, but hallucinated citations and opaque reasoning remain persistent problems [1, 8].

Retrieval-augmented generation. RAG [10] reduces hallucination by conditioning LLM output on retrieved documents rather than parametric memory. Hybrid search combining dense and sparse retrieval tends to outperform either method alone, and chunk-level metadata is important for verifiable citations [7]. Automated evaluation frameworks such as RAGAS [13] have been developed to assess RAG pipeline quality.

Gap. No existing open-source legal AI system, to our knowledge, combines retrieval-based grounding with symbolic consistency checking and cryptographic audit logging. EALAI integrates all three as first-class architectural components.

3 System Architecture

Figure 1 shows the eight-stage pipeline. Stages 1–5 form the RAG core; stages 6–8 are the EALAI additions.

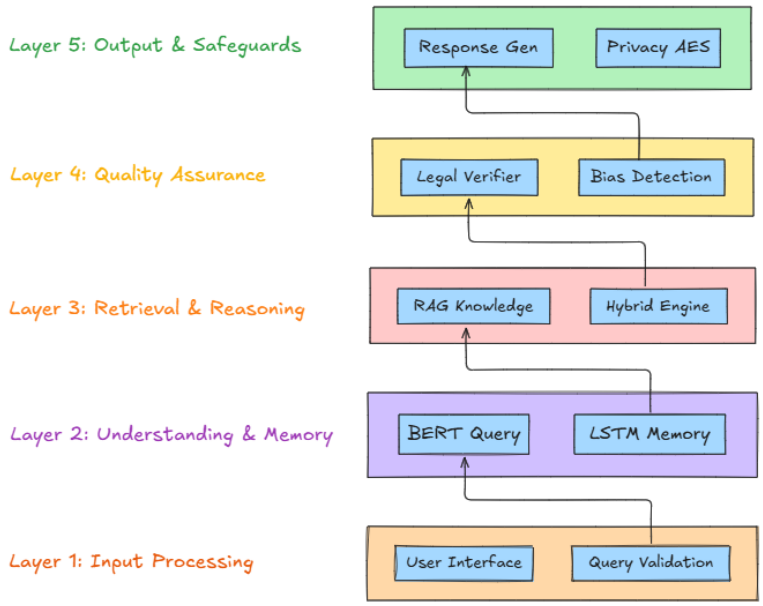

Fig. 1. EALAI eight-stage pipeline from user input to audited response.

3.1 Retrieval Pipeline (Stages 1–3)

Preprocessing. A keyword-based intent classifier identifies legal domains (employment, contract, PDPA) and spaCy extracts named entities (locations, persons, dates). Both outputs are injected into the prompt.

Corpus. 48 Singapore statutory PDFs are chunked using paragraph-boundary splitting with token-aware sizing (target 500–1000 tokens, 15% overlap) and indexed in ChromaDB with `all-MiniLM-L6-v2` embeddings (384 dimensions).

Retrieval modes. The system supports three modes:

- **BM25 Okapi** [11] ($k_1 = 1.5$, $b = 0.75$) for keyword retrieval.
- **Dense search** via cosine similarity over chunk embeddings.
- **Hybrid (default):** both run independently and are merged using Reciprocal Rank Fusion [12] with $k = 60$: $\text{score}(d) = \sum_i \frac{1}{k+\text{rank}_i(d)}$.

3.2 Generation and Formatting (Stages 4–5)

A Jinja2 template injects conversation history, retrieved chunks (with source citations), intent/entity slots, and output format instructions into the prompt. The LLM (`llama-3.1-8b-instant` via Groq, $T = 0.1$) must respond in five labelled sections: Applicable Law, Reasoning, Answer, Limitations, and Disclaimer, plus a confidence tag. A response formatter parses these sections via regex-based header detection.

3.3 Symbolic Consistency Engine (Stage 6)

The `SingaporeLegalEngine` holds 20 IF-THEN rules encoded as Python dataclasses, covering the Employment Act 1968 (rules 1–8), Contract Law (9–12), and PDPA (13–20). Each rule carries a trigger condition, statute name, and section reference. The `check_consistency()` function compares the matched statute against the LLM's stated Applicable Law, producing a consistency flag (`True`/`False`/`None`).

3.4 Audit Logger and Citation Checker (Stages 7–8)

Every interaction is appended to an audit log as a JSON object. A SHA-256 hash is computed over all field values (sorted by key, pipe-joined) for tamper detection. A citation checker extracts `[Act Name, Section X]` patterns from the LLM output via regex and verifies each against retrieved chunk texts, producing a `citation_accuracy` score in $[0, 1]$.

4 Evaluation

4.1 Benchmark

We constructed a benchmark of 200 questions (74 employment, 62 contract, 64 PDPA) with gold-standard answers, statute references, and chunk IDs. For the refusal calibration experiment, 100 additional out-of-scope (OOS) queries were annotated across 8 categories (wrong jurisdiction, non-legal, fictional case, etc.).

4.2 Retrieval Quality

Table 1 reports Precision@k and Hit@k for $k \in \{3, 5\}$ across 142 scoreable questions (58 excluded for non-statute gold references).

Table 1. Retrieval benchmark ($n = 142$).

Retriever	P@3	Hit@3	P@5	Hit@5
BM25	0.279	50.7%	0.249	56.3%
Dense	0.289	57.8%	0.249	65.5%
Hybrid (RRF)	**0.322**	**58.5%**	**0.293**	**72.5%**

Hybrid retrieval consistently outperforms either retriever alone. The largest gap is at Hit@5 (72.5% vs. 65.5% for dense), confirming that RRF fusion recovers relevant chunks missed by individual retrievers.

4.3 End-to-End Evaluation

Four retrieval conditions were evaluated across all 200 questions. Following the approach of automated RAG evaluation [13], each response was scored for hallucination rate and citation accuracy (Table 2, Figure 2).

Table 2. End-to-end results across four retrieval conditions ($n = 200$).

Retriever	Hallucination	Citation Acc.	Latency
None (baseline)	42%	0.580	30.7 s
BM25	22%	0.784	34.3 s
Dense	**20%**	**0.800**	32.5 s
Hybrid	36%	0.645	33.1 s

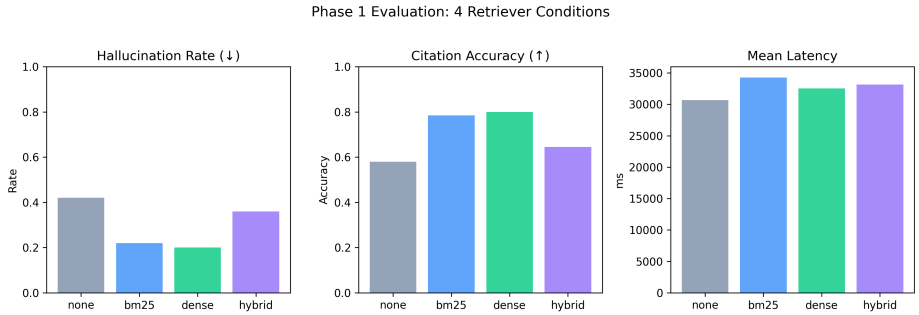

Fig. 2. Hallucination rate, citation accuracy, and latency across four conditions. Latency includes Groq API round-trip time.

Dense retrieval achieves the best end-to-end quality: 20% hallucination rate and 0.800 citation accuracy. The key finding is that hybrid retrieval, despite winning on raw retrieval recall (Table 1), underperforms on citation accuracy (0.645 vs. 0.800). The RRF-merged context introduces topical diversity that the LLM cannot reliably attribute, producing a dissociation between retrieval coverage and response quality.

4.4 Refusal Calibration

The refusal mechanism rejects queries where the maximum retrieval score falls below a threshold θ. We swept $\theta \in \{0.20, 0.25, \ldots, 0.70\}$ using 100 OOS and 100 in-corpus queries per threshold (Figure 3).

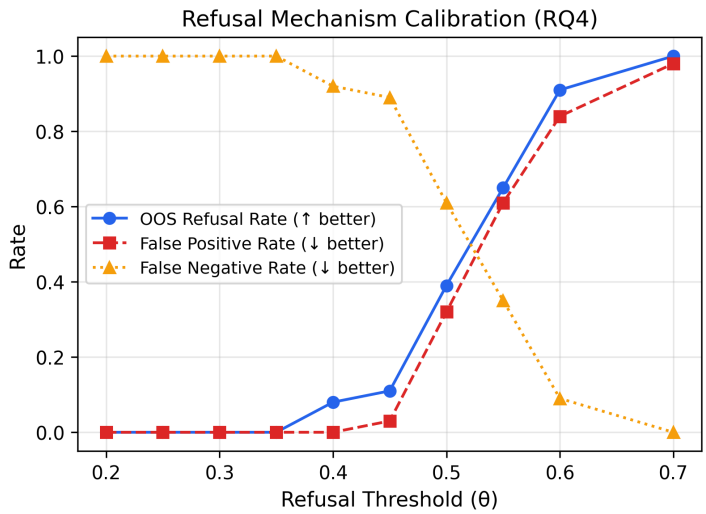

Fig. 3. Refusal calibration sweep. The false positive rate jumps from 3% to 32% between $\theta = 0.45$ and $\theta = 0.50$.

The curve shows a sharp cliff between $\theta = 0.45$ and $\theta = 0.50$: the false positive rate jumps from 3% to 32% for only a 28pp gain in OOS recall. The recommended operating point is $\theta = 0.45$.

5 Discussion

5.1 Key Findings

RAG substantially reduces hallucination. Dense retrieval cuts the hallucination rate from 42% to 20%, a 22pp reduction, confirming that grounding LLM output in verified statutory text is effective even with a small (8B parameter) model.

Retrieval coverage $\neq$ response quality. Hybrid retrieval achieves the best Hit@5 (72.5%) but the worst citation accuracy among RAG conditions (0.645). This suggests that the LLM's ability to correctly attribute claims degrades when the context contains topically diverse chunks from multiple retrieval signals. For downstream citation quality, a focused, high-precision context appears preferable to a broad, high-recall one.

Refusal threshold has a sharp cliff. The precision-recall tradeoff for the refusal mechanism is non-linear, with a single 0.05 threshold step producing a tenfold increase in false positives. This highlights the need for careful calibration rather than relying on default values.

5.2 Limitations

- **Automated evaluation only.** Citation accuracy is computed via regex matching, not expert annotation. Subtle hallucinations (correct statute, wrong interpretation) may be missed.
- **Small model.** All experiments use Llama-3.1-8b-instant. Results may not generalize to larger models that handle diverse contexts differently.
- **Corpus scope.** The 48-PDF corpus covers three domains within Singapore law. The system cannot assess its own blind spots within covered domains.
- **Rule coverage.** The 20-rule symbolic engine covers a subset of scenarios. Queries between rules receive no consistency check.
- **No user study.** Perceived explainability and response clarity have not been measured with human participants.

6 Conclusion

We presented EALAI, a legal AI system combining hybrid RAG retrieval, a 20-rule symbolic consistency engine, and SHA-256 tamper-evident audit logging. Evaluation on a 200-question Singapore law benchmark shows that retrieval-augmented generation reduces hallucination by 22 percentage points and that retrieval coverage does not automatically translate into citation accuracy. The refusal calibration sweep identifies a sharp precision cliff that underscores the importance of threshold tuning.

Future work includes expanding the symbolic rule base, conducting a user study with legal aid recipients, developing a composite Explainability Index metric, and evaluating on larger language models and additional jurisdictions.

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